

GPUs and cooling system

1. Open your own laptop or desktop, identify whether it uses an integrated GPU, a discrete GPU, or both, and write down the GPU model(s) you find.

->You can complete this practical by checking your own laptop or desktop. I cannot directly access your computer to identify its GPU.

Steps to Identify Your GPU

Method 1: Using Windows Task Manager

1. Press **Ctrl + Shift + Esc** to open Task Manager.
2. Click **Performance**.
3. Look for **GPU 0, GPU 1**, etc.
4. Write down the GPU model shown.

Method 2: Using Device Manager

1. Right-click the **Start** button.
2. Select **Device Manager**.
3. Expand **Display adapters**.
4. Note the GPU model(s).

Assignment Format

Item	Your Result
GPU Type	Integrated / Discrete / Both
GPU Model	_____
GPU Model 2 (if present)	_____

Example

If a laptop shows **Intel UHD Graphics** and **NVIDIA GeForce RTX 3050** under Display adapters:

- **GPU Type:** Both integrated and discrete

- **Integrated GPU:** Intel UHD Graphics
- **Discrete GPU:** NVIDIA GeForce RTX 3050

Note: If you see only one GPU, it may be either an integrated or discrete GPU depending on the model.

2. List all cooling system components you can physically see in your system (fans, heatsinks, vents, etc.), and briefly describe the function of each.

->Cooling System Components in a Computer

You can physically inspect your laptop or desktop and use this table for your assignment. Only list the components that are actually visible in your system.

Cooling Component	Function
CPU Fan	Moves air across the CPU heatsink to remove heat from the processor.
CPU Heatsink	Absorbs heat from the CPU and transfers it to the surrounding air.
Case/Chassis Fans	Circulate cool air into the case and push hot air out.
GPU Fan	Cools the graphics processor by moving air through the GPU heatsink.
GPU Heatsink	Transfers heat away from the GPU and helps maintain safe operating temperatures.
Air Vents	Allow cool air to enter and hot air to escape from the computer.
Heat Pipes	Transfer heat from hot components such as the CPU/GPU to a heatsink. These are especially common in laptops.
Thermal Paste	Fills tiny gaps between the CPU/GPU and heatsink, improving heat transfer. It is usually not visible without removing the heatsink.

Example Observation

If you are examining a typical desktop PC, you might write:

Cooling components observed: CPU fan, CPU heatsink, case fans, GPU fan, GPU heatsink, and ventilation openings. The fans move air through the system, while the heatsinks absorb and dissipate heat from the CPU and GPU. The vents provide a path for cool air to enter and hot air to leave.

Tip: Do not remove the CPU or GPU heatsink just to complete this practical. Some cooling components, such as thermal paste, may not be physically visible during a normal inspection.

3. Download and use a free CPU/GPU temperature monitoring tool (like HWMonitor or Speccy), then record the idle and under-load temperatures of your CPU and GPU.
Hint: Run a browser video or simple game for 5 minutes to simulate load.

->You can complete this practical using **HWMonitor** or another temperature-monitoring tool. Since I cannot access your computer's sensors, you should record the actual temperatures from your system rather than using made-up values.

Procedure

1. Install and open **HWMonitor**.
2. Leave the computer idle for about **5 minutes** without running demanding applications.
3. Record the **CPU temperature** and **GPU temperature**.
4. Start a browser video or a simple game and run it for about **5 minutes**.
5. Check HWMonitor again and record the temperatures under load.
6. Take a screenshot of the HWMonitor readings for your assignment.

Temperature Record Table

Condition	CPU Temperature	GPU Temperature
Idle	____ °C	____ °C
Under Load (5 minutes)	____ °C	____ °C

Example Format

The numbers below are only an example—use your own measured temperatures.

Condition	CPU	GPU
Idle	45°C	40°C
Under Load	72°C	68°C

Observation

The CPU and GPU temperatures increased when the computer was under load because both components were performing more calculations. The cooling system, including fans and heatsinks, helps remove the additional heat and keep the components within their normal operating temperature range.

4. Create a thermal management checklist for a gaming PC, including steps for checking thermal paste, cleaning fans, and monitoring temperatures

-> Here is a practical **thermal management checklist for a gaming PC**:

Gaming PC Thermal Management Checklist

1. Check Temperatures

- ☐ Monitor CPU temperature at idle.
- ☐ Monitor GPU temperature at idle.
- ☐ Check CPU and GPU temperatures while gaming.
- ☐ Watch for unusual temperature increases or thermal-throttling warnings.
- ☐ Record temperatures periodically to identify changes over time.

2. Clean Fans and Vents

- ☐ Shut down the PC and disconnect the power before cleaning.
- ☐ Clean dust from case fans and ventilation openings.
- ☐ Clean dust from CPU and GPU cooling areas.
- ☐ Use compressed air carefully to remove dust.
- ☐ Make sure fans can spin freely and are not blocked by cables or dust.
- ☐ Keep the PC case in a well-ventilated location.

3. Check Thermal Paste

- ☐ Check the CPU cooler if temperatures remain unusually high.
- ☐ Inspect the thermal paste when the cooler is removed.
- ☐ Replace old or dried thermal paste when necessary.
- ☐ Apply an appropriate amount of new thermal paste according to the cooler manufacturer's instructions.
- ☐ Reinstall the CPU cooler evenly and securely.

4. Check Airflow

- ☐ Make sure intake and exhaust fans are positioned correctly.
- ☐ Keep front and rear ventilation areas unobstructed.
- ☐ Organize cables so they do not block airflow.
- ☐ Check that all cooling fans are working properly.

5. Regular Maintenance

- ☐ Check temperatures after cleaning or replacing thermal paste.
- ☐ Clean the PC periodically, especially in dusty environments.
- ☐ Listen for unusual fan noises.
- ☐ Replace a malfunctioning fan if necessary.
- ☐ Keep temperature-monitoring software available for troubleshooting.

Safety Reminder

Always turn off and unplug the PC before opening the case. Avoid touching components unnecessarily, and follow the manufacturer's instructions when removing a CPU cooler or applying thermal paste.

Conclusion

Good thermal management helps a gaming PC maintain stable performance and reduces the risk of overheating. Regular cleaning, proper airflow, appropriate thermal paste, and temperature monitoring are the main parts of an effective cooling routine.

5. Given these three combinations: (A) Intel i5 CPU + GTX 1650 GPU, (B) Ryzen 9 CPU + RTX 4080 GPU, (C) Intel i3 CPU with integrated graphics, recommend an appropriate cooling solution (air or liquid) for each and justify your choice based on TDP values.

-> Here is a suitable cooling recommendation based on the typical TDP/power levels of these components:

Combination	Typical TDP / Power Level	Recommended Cooling	Justification
A: Intel i5 + GTX 1650	CPU ~65–125 W; GPU ~75 W	Air cooling	The i5 and GTX 1650 produce moderate heat. A good CPU air cooler plus normal case airflow is generally sufficient and is more affordable than liquid cooling.
B: Ryzen 9 + RTX 4080	CPU ~120–170 W depending on model; GPU ~320 W	Liquid cooling	This combination produces substantially more heat. A 240/280/360 mm AIO liquid cooler is appropriate for the Ryzen 9, combined with strong case airflow for the RTX 4080.
C: Intel i3 + Integrated Graphics	CPU typically ~35–65 W; integrated graphics uses the CPU's thermal budget	Basic air cooling	Heat output is relatively low, so the stock Intel cooler or a basic tower air cooler is normally adequate. Liquid cooling would provide little practical benefit.

Conclusion

- **A → Air cooling:** moderate TDP, simple and cost-effective.
- **B → Liquid cooling:** high CPU + GPU power and significantly greater heat output.
- **C → Basic air cooling:** low TDP and integrated graphics make liquid cooling unnecessary.

Note: Exact TDP varies by the specific Intel i5/i3 and Ryzen 9 model. Also, the GPU has its own cooler; CPU cooling does not directly cool the RTX 4080.

